4,212,2 Macroeconomics III
Exercises and Independent Studies

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Outline

- Unemployment: Stylized Facts (Exercise 6.1)
- The Efficiency Wage Model (Exercise 6.2)

Unemployment: Stylized Facts

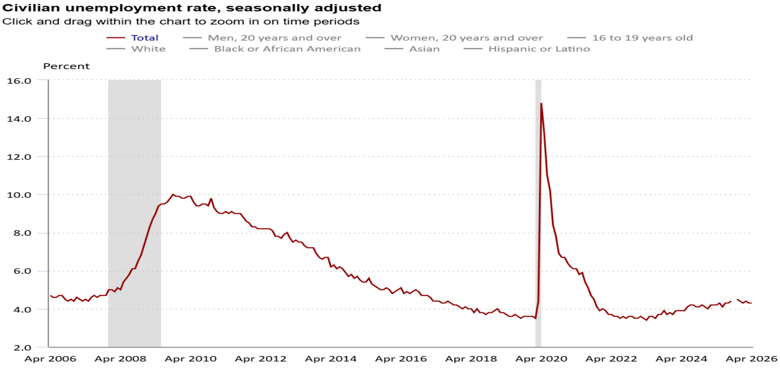
- How is the rate of unemployment defined and measured by the U.S. Bureau of Labor Statistics?
- Each month, the U.S. Bureau of Labor Statistics (BLS) announces the total number of employed and unemployed people in the U.S. for the previous month, along with many characteristics about them.
- Because unemployment insurance goes only to people who have applied for such benefits and it is impractical to count every unemployed person each month, the government conducts a monthly survey called the Current Population Survey (CPS) to measure unemployment in the country.
- The CPS has been conducted in the U.S. every month since 1940. About 60,000 eligible households are in the sample (about 110,000 individuals) each month. The CPS sample is selected to represent the entire civilian noninstitutional population of the U.S.

Unemployment: Stylized Facts

- How is the rate of unemployment defined and measured by the U.S. Bureau of Labor Statistics?
- The basic concepts involved in identifying the employed and unemployed are quite simple:
  - People with jobs (part-time and temporary work, as well as regular full-time, year-round employment) are employed.
  - People are classified as unemployed if they do not have a job, have actively looked for work in the prior 4 weeks, and are currently available for work.
  - The labor force comprises the employed and the unemployed. Those who have no job and are not looking for one are counted as not in the labor force.
  - The survey excludes people living in institutions (for example, a correctional institution or residential nursing or mental health care facility) and those on active duty in the Armed Forces.
  - The unemployment rate is computed as the percentage of the labor force that is unemployed.

Unemployment: Stylized Facts

- Explain in one paragraph the evolution of the unemployment rate over the period Apr. 2006- Apr. 2026.



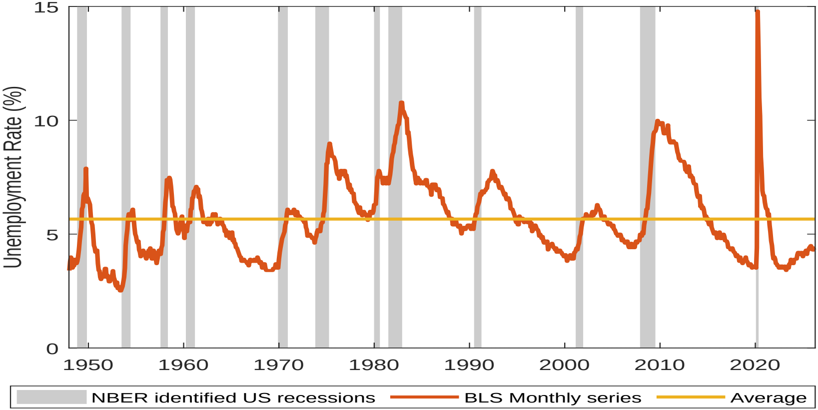
Source: U.S. Bureau of Labor Statistics.

Unemployment: Stylized Facts

- There are short-run (cyclical) fluctuations of unemployment around the natural rate, sometimes large (14.8 % in April 2020)!
- These are mainly the patterns of cyclical unemployment: temporary changes in unemployment due to economic expansions or recessions.
- However, the spike in unemployment in 2020 is due to the public health response to the coronavirus pandemic, an economic shock itself, and is not the usual cyclical unemployment associated with a recession.

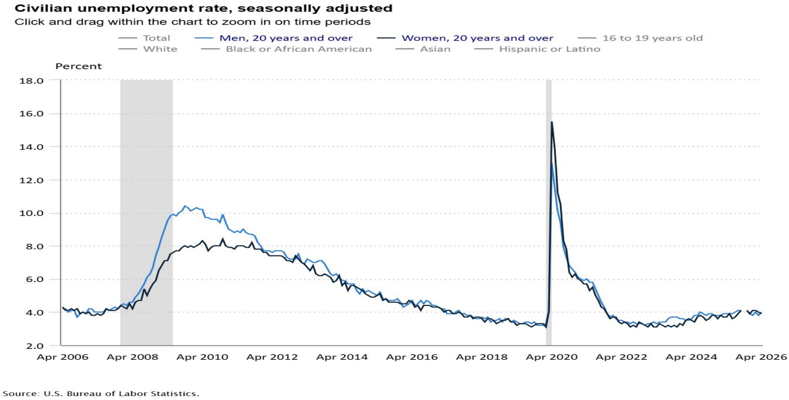
Unemployment: Stylized Facts

- The average unemployment rate from Jan. 1948 to Apr. 2026 is 5.66%.



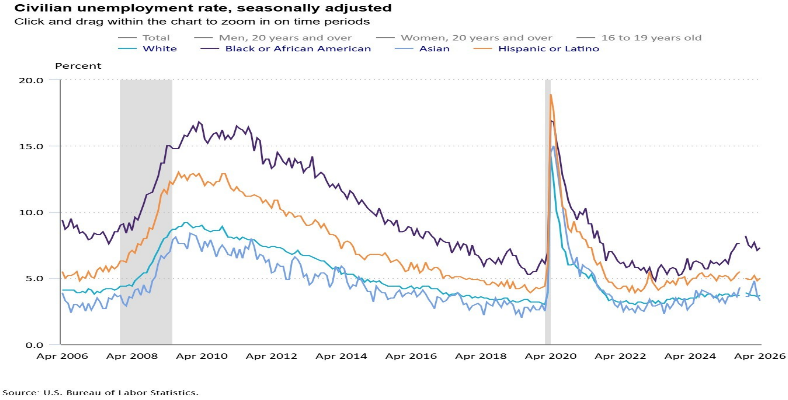
Unemployment: Stylized Facts

- Variation in unemployment rates by gender:



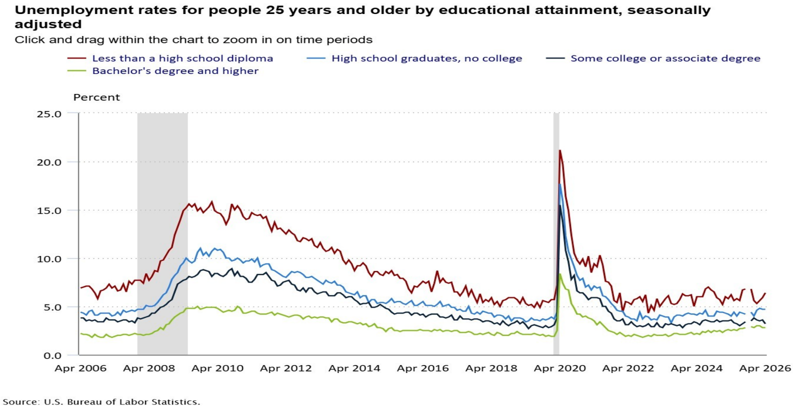
Unemployment: Stylized Facts

- Variation in unemployment rates by ethnicity:



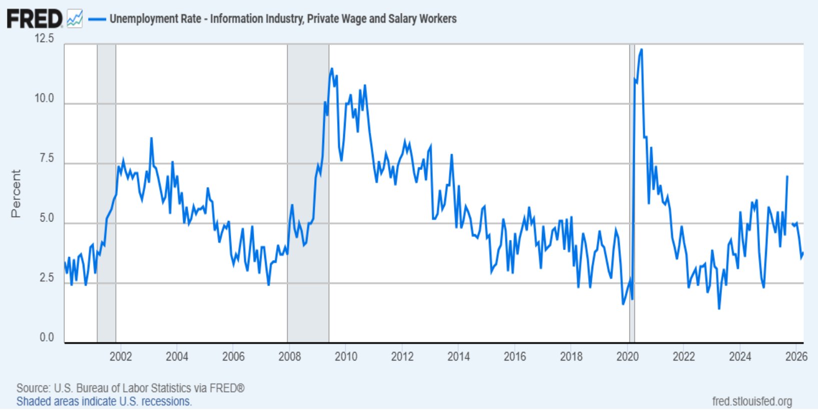
Unemployment: Stylized Facts

- Variation in unemployment rates by educational attainment:



Unemployment: Stylized Facts

- Unemployment in the information industry:



The Efficiency Wage Model

The efficiency wage model with one firm, whose profits are $zR(a(w')L) - w'L$.

- The firm chooses both w' and L in order to maximize profits:

$$\max_{w',L} zR(a(w')L) - w'L$$

- The first order condition w.r.t w':

$$zR'(a(w')L)a'(w')L = L \tag{1}$$

- The first order condition w.r.t L :

$$zR'(a(w')L)a(w') = w' \tag{2}$$

The Efficiency Wage Model

- Divide equation (1) by equation (2) to get:

$$\frac{a'(w')L}{a(w')} = \frac{L}{w'} \rightarrow \frac{a'(w')w'}{a(w')} = 1 \tag{3}$$

- This is equivalent to:

$$\frac{\partial a / a}{\partial w' / w'} = 1$$

- This implies that the elasticity of labor productivity w.r.t. the real wage is 1 (one percent increase in w' exactly gives a one percent increase in a).

The Efficiency Wage Model

- The Solow condition: If the labor productivity of workers in a firm depends positively on the worker's pay through an efficiency function, $a(w)$, then an (interior) optimal wage rate w of the firm depends only on this efficiency function, and w^* is given by the condition that the elasticity of productivity w.r.t. real wage is 1, as in equation (3).
- To derive the optimal wage w^*, we use that the efficiency function is

$$a = (w - v)^\eta, \qquad 0 \leq \eta < 1 \tag{4}$$

where v is the employees' outside option.[1]
- We introduce equation (4) in equation (3) to obtain that

$$w^* = \frac{v}{1-\eta}. \tag{5}$$

[1]The efficiency function given is for $w \geq v$. For $w < v$, $a = 0$.

The Efficiency Wage Model

$$w^* = \frac{v}{1-\eta}.$$

- How does the optimal wage depend on v and η?
- The optimal wage is a mark-up $\frac{1}{1-\eta} > 1$ over the outside option (alternative or normal income v).
- The optimal wage is determined relative to v (if v increases, so does w). A high outside option implies that the firm must offer a high wage to attract workers.
- The optimal wage increases with η. The higher η the larger the productivity effect of higher wages (a is more elastic to the gap between w and v).

The Efficiency Wage Model

Using the fact that the outside option is

$$v = (1-u)w + ucw, \qquad 0 < c < 1,$$

where u is the unemployment rate, c is the replacement ratio, and w is the general real wage level, we can rewrite the optimal wage function as

$$w' = \frac{(1-u)w + ucw}{1-\eta} = \frac{1-u(1-c)}{1-\eta}w.$$

Using the equilibrium condition $w = w'$, we can derive the equilibrium rate of unemployment u^*

$$1 = \frac{1-u(1-c)}{1-\eta} \to u = u^* = \frac{\eta}{1-c}.$$

The Efficiency Wage Model

$$u^* = \frac{\eta}{1 - c}.$$

- How do the parameters η and c affect u^*?
- A higher η, as well as a higher c, increase the equilibrium rate of unemployment.
- The higher real wage elasticity of effort or productivity η, the higher the wage and thus labor productivity of workers, implying a lower demand for workers.
- The higher c, the higher the unemployment benefits, and thus the less incentivized workers are to return to work.
- Note that for the expression for u^* to be meaningful one must have that $u^* < 1$, or $\eta < 1 - c$, which is a reasonable condition.

The Efficiency Wage Model

Assume that a proportional labor income tax is introduced. How does this impact the outside option equation?

$$v = (1-u)(1-\tau)w + ucw,$$

where τ is the labor income tax.

- The efficiency function also changes:

$$a = (w'(1-\tau) - v)^{\eta}$$

- This implies that

$$w' = \frac{v}{(1-\eta)(1-\tau)}.$$

- The tax on labor income increases the optimal wage rate.
- The tax reduces the net benefit of being employed in the firm given w' and v. The firm, therefore, has to set a higher w' in order to induce (optimal) effort.

The Efficiency Wage Model

- Using the equilibrium condition $w = w'$, we can derive the equilibrium rate of unemployment u^*

$$u = u^* = \frac{\eta(1-\tau)}{(1-\tau)-c}.$$

- To find the effect of taxation on u^* we differentiate w.r.t. τ:

$$\frac{\partial u^*}{\partial \tau} = \frac{\eta c}{(1-\tau-c)^2} > 0$$

- Thus, a higher tax rate increases equilibrium unemployment. The intuition is that the net benefit from being employed is reduced which induces firms to raise w. In equilibrium u^* must rise.

The Efficiency Wage Model

How does your answer change if the unemployment benefits are taxed at the same rate?

$$v = (1-u)(1-\tau)w + ucw(1-\tau).$$

Performing the same steps as before, we get

$$u^* = \frac{\eta}{1-c}.$$

- This time the tax on income has no effect on the natural rate of unemployment.
- When unemployment benefits are taxed at the same proportional rate as labor income the taxation affects the net benefit from employment and the outside option the same way and thus does not induce firms to raise wages.